

Lesson Objective

Interpret and represent patterns when multiplying by 10, 100, and 1,000 in arrays and numerically.

Materials

(T/S) Thousands Place Value Chart (S) Personal white board

PROBLEM

- 1 **Represent 4 times 10, 100, and 1,000 with place value disks and find the products.**

4×10
 4×100
 $4 \times 1,000$

Answers: 40, 400, 4,000

TEACHER GUIDANCE

Ask students to represent each problem on a place value chart and find each product.

Then, **model** how to draw place value disks and use arrows to represent the multiplication.

- 2 **Represent multiplication by 10, 100, and 1,000 by drawing place value disks and find the products.**

7×10
 2×100
 $8 \times 1,000$

Answers: 70, 200, 8,000

Students work independently or with a partner, continuing to use concrete disks as necessary.

Ask students to name patterns they notice when multiplying by 10, 100, and 1,000.

- 3 **Multiply a two-digit number by 10.**

34×10

Answer: 340

Model for students. Establish that the pattern holds when we multiply by two-digit numbers, also.

- 4 **Multiply a two-digit number by a multiple of 10.**

57×100
 14×100

Answers: 5,700, 1,400

Students work independently or with a partner.